

Musculoskeletal Diseases

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Small mammals suffer from a range of musculoskeletal diseases, many being the same as in other domestic mammals. We will only cover the most important, common, and some other selected diseases of small mammals. Fractures will be covered in a discussion session, we'll just mention the special case of the chinchilla.

RABBITS

Osteoarthritis/Spondylosis

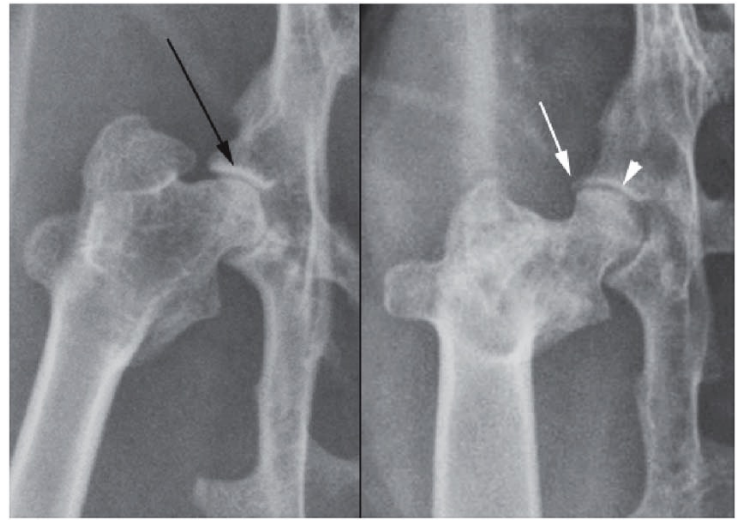
Osteoarthritis also known as degenerative joint disease is an age-related change that is **common** in rabbits (over 40% prevalence). **Spondylosis (spinal osteoarthritis) is also very common** in rabbits.

Osteoarthritis is **more common in older animals and non-dwarf breeds** seem to be somewhat predisposed to it. In the appendicular skeleton, **it is most commonly diagnosed in the hip and knee joints**. No particular association with gender has been shown.

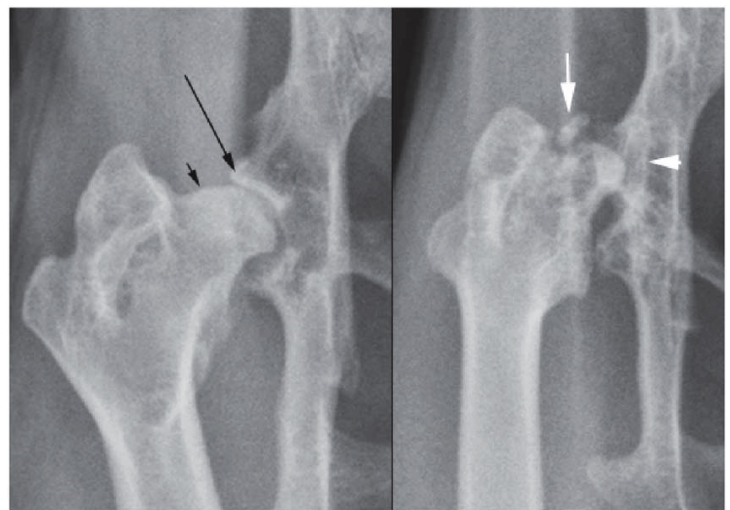
Clinical signs are mainly related to pain and joint function and include stiffness of gait, lameness, decreased range of motion, crepitus and other signs of chronic pain.

Arthritic lesions can be diagnosed on radiographs. Cytology and culture for painful joints may be indicated to differentiate from another disease process such as septic arthritis or neoplasia.

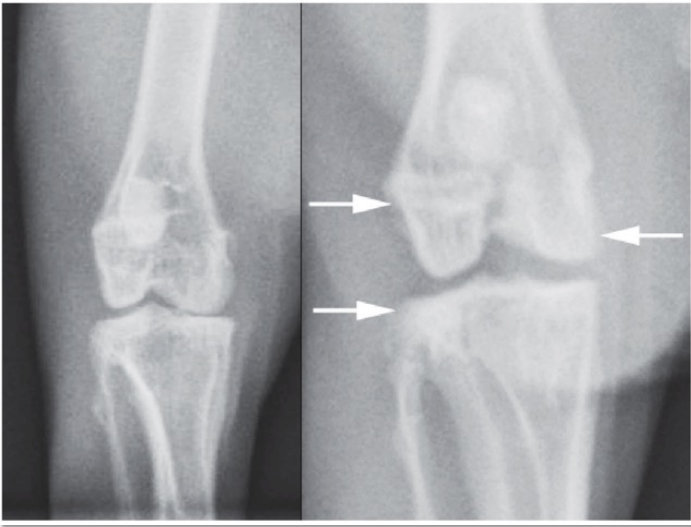
Treatment mainly consists of weight loss in overweight rabbits and optimizing housing with nonslip, soft and clean surfaces. Chronic pain medications such as with **NSAIDS** (for instance **meloxicam** 0.5 mg/kg PO q12h) are often useful. There is also a wide array of other therapeutics that may be tried, much like in other species, such as additional pain medications (gabapentin, tramadol), chondroprotectors: chondroitin sulfate, polysulfated glycosaminoglycan, and omega 3 fatty acids supplementation, and physical therapy.



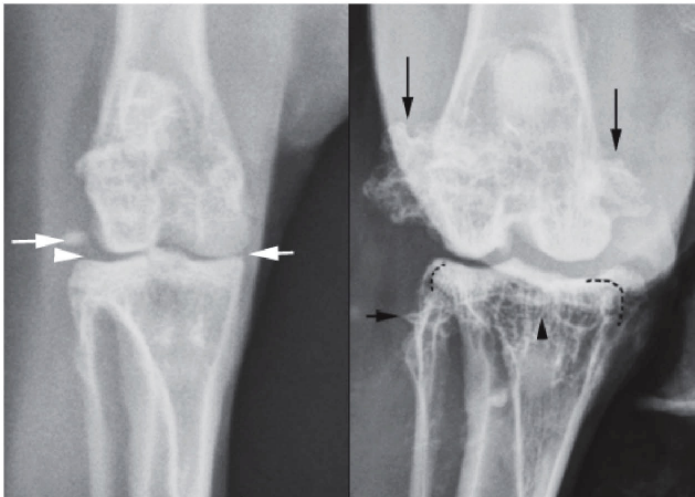
Radiographic images of hip joints with increasing degrees of osteoarthritis severity. Left: healthy hip joint, normal joint space (arrow). Right: mild osteoarthritis, minimal periarticular new bone formation (arrow) and regional joint space narrowing (arrowhead) are visible.



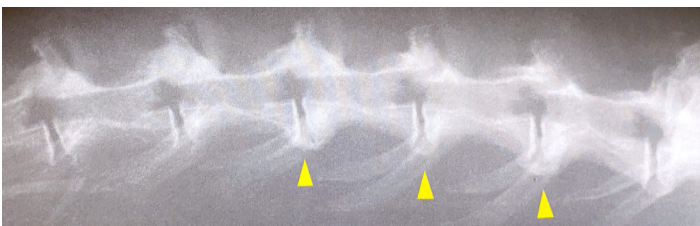
Left: moderate osteoarthritis, mild subchondral bone sclerosis (long arrow) and remodeling of the femoral head and neck associated with new bone formation (short arrow). Right: marked osteoarthritis. Pronounced coxofemoral subluxation, a large osseous body in the joint space (arrow) and remodeling of femoral head and neck and acetabulum (arrowhead).



Radiographic images of knee joints with increasing degrees of osteoarthritis severity. Left: normal knee joint. Right: mild osteoarthritis. Minimal periarticular new bone formation arising from the distal femur and proximal tibia (arrows) is visible.



Left: moderate osteoarthritis. Minimal periarticular new bone formation involving the medial aspect of the proximal tibia (right arrow), meniscal mineralization (left arrowhead) and a separate osseous body in the joint space (left arrow) are visible. Right: marked osteoarthritis. Pronounced periarticular new bone formation involving the distal femur (long arrows) and proximal tibia (dashed lines), architectural remodeling of subchondral bone in the proximal tibia (short arrow) and subchondral sclerosis of the lateral and medial femoral condyles (arrowhead) are visible.



Spondylosis in a rabbit. Multiple "bridging" between vertebral bodies are seen (yellow arrows).

Splay Leg Syndrome

A number of disorders characterized clinically by complete abduction of one or more legs and the inability to assume a normal standing position are described by the term "splay leg". This condition is seen in young rabbits from neonates to juveniles that are several months old. **Young rabbits 3 to 4 weeks of age are most commonly affected.** The condition may be unilateral or bilateral and may affect the anterior, posterior or all four limbs in both sexes. The degree of splaying may be relatively mild, or so severe that the animal is completely unable to walk. **Splay leg is also seen in geriatric rabbits. The hind limbs are more commonly affected in young rabbits whereas the front limbs are more affected in senior rabbits.**



Rabbits housed on a **smooth slippery flooring or with inappropriate diet** may be predisposed. This condition may also have a genetic component in a simple autosomal recessive pattern. Finally obesity and osteoarthritis may be contributing factors.

If caught early, splay leg may be corrected by **splinting and hobbling** affected legs in addition to improving flooring and diet. Young animals with this disorder should be prevented from breeding because of its heritability.

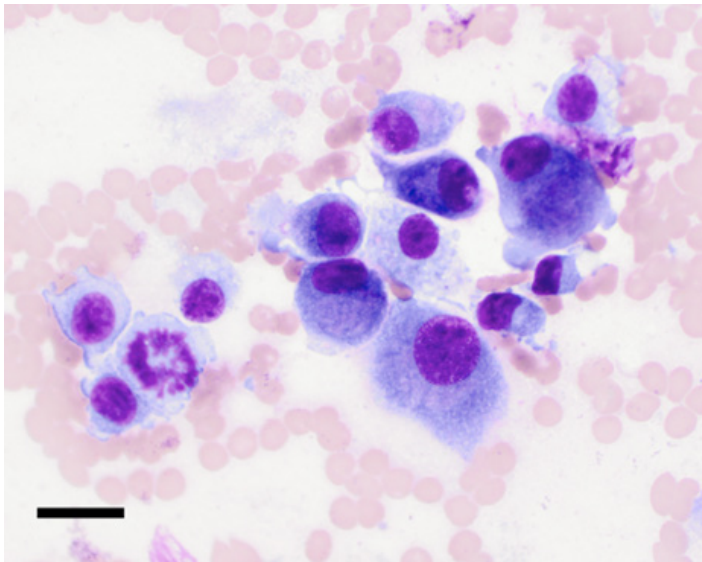
Osteosarcoma

Osteosarcoma is **relatively rare** in rabbits, but remains an important differential for bone abnormalities and masses connected to the bones (such as odontogenic abscesses). Most reports pertain to osteosarcoma arising from the axial skeleton, but appendicular osteosarcoma can also occur. Mandibular osteosarcoma is also an important differential for masses and lesions

of the jaw. Both osteoblastic or osteoclastic lesions can be seen on imaging. Clinical signs vary depending on which bone is affected, but often pain and lameness are present. Bone or mass biopsy or cytology are necessary for diagnosis.



Craniocaudal radiographic views of the left stifle of a domestic rabbit approximately 4 months after the onset of lameness. Mass appears as an ovoid mineralized structure lateral to the left tibia and fibula



Fine needle aspirate of an appendicular osteosarcoma in a domestic rabbit. Scale bars 20µm. Round to spindlyloid neoplastic cells are noted. Anisocytosis and anisokaryosis are marked. A mitotic figure is seen.

Once diagnosed, limb amputation, chemotherapy, and radiation treatment can be considered.

FERRETS

Myofasciitis

Disseminated idiopathic myofasciitis affects young ferrets, typically less than 1-1.5 years of age. No cause has been identified so far; a possible link to vaccination with distemper vaccine has been suggested but remains unproven. Overall, an immune-mediated process of some kind is suspected.

Ferret with myofasciitis presents with relatively non-specific signs. With that said, they are often lethargic, appear **painful and are pyrexia** (fever is typically fairly common with this disease). Other signs may include hindlimb paresis, difficulty swallowing and drinking, bruxism, and a painful palpation of the muscles. Progression varies with individuals declining rapidly and others having a more chronic disease course.

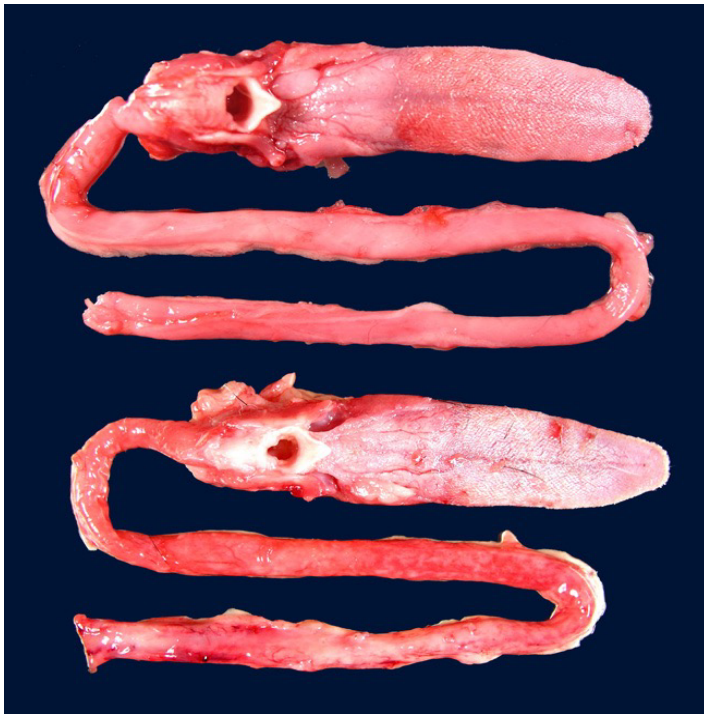
On bloodwork, a **severe neutrophilic leukocytosis** is commonly found as well as an anemia. Interestingly, **CK is rarely elevated**, despite being a muscle marker. Other enzymes (such as AST) and biochemical parameters are likewise not impacted. Nothing is typically seen on diagnostic imaging, which should still be performed to rule out other causes for the pyrexia.

The disease may be confirmed by taking several biopsies from different sites, especially atrophied muscles. The vastus lateralis, cranial tibial, and triceps brachii are common biopsy targets. If muscle biopsies are taken, a **moderate to severe generalized suppurative to pyogranulomatous inflammation will be seen in muscles and fascia** (hence the name for the disease).

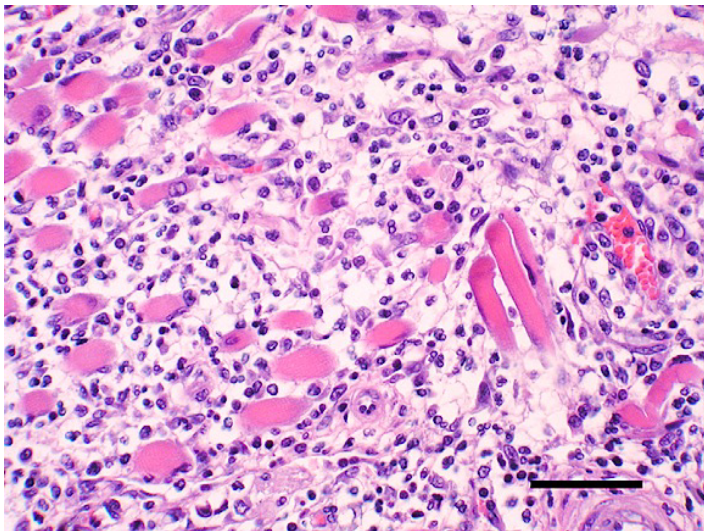
On necropsy, lesions can also be found in the **muscles of the esophagus**, actually in the entire esophagus, which is somewhat pathognomonic for this disease. Other lesions that may be seen are pneumonia, and splenic and bone marrow myeloid hyperplasia.



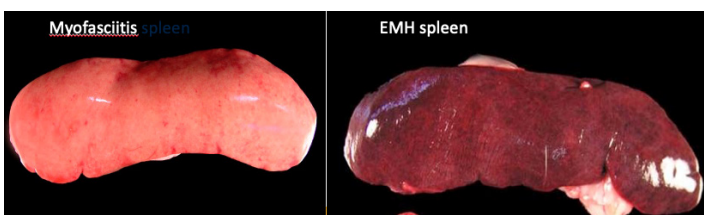
Left: myofasciitis, right: normal



Esophagus of ferrets. Note full length mottling of the esophagus, shrinkage of the tongue and lingual hyperkeratosis of ferret with polymyositis (bottom) compared with age matched ferret that died of unrelated disease (top).



Ferret, myofasciitis. Muscular tunic of esophagus. Note myofiber disarray owing to infiltrate of large numbers of neutrophils (H&E stain, bar 5 160 mm).



Treatment may be unrewarding and consists of **immunosuppressive medications** such as a combination of prednisolone, cyclophosphamide, and chloramphenicol in addition to supportive care. The prognosis is grave for most cases.

GUINEA PIGS

Vitamin C Deficiency

Guinea pigs are **unable to synthesize vitamin C endogenously and rely on dietary sources**. They lack the gene for L-gulonolactone oxidase, which prevents the conversion of L-gulonolactone into L-ascorbic acid. If not enough vitamin C is provided in the diet, guinea pigs will develop clinical signs related to deficiency, which is also called **scurvy**. Young and growing animals are more likely to develop deficiencies. Guinea pigs require at least 10-25 mg/k/day in their diet.

Deficiencies lead to **defective collagen synthesis** as well as laminin and elastin, which in turn lead to compromise blood vessels and cartilage. Clinically, this manifests by **gingival bleeding, loose teeth as well as malocclusion, swollen and painful joints** (see pictures below: swollen stifle, periarticular hemorrhage and **enlargement of costochondral junction of the ribs**), rough hair coat, diarrhea and immunosuppression.



Common differentials include osteoarthritis, rickets (metabolic bone disease), anticoagulant rodenticides (because of bleeding), other coagulopathy and many other as some signs are non-specific.

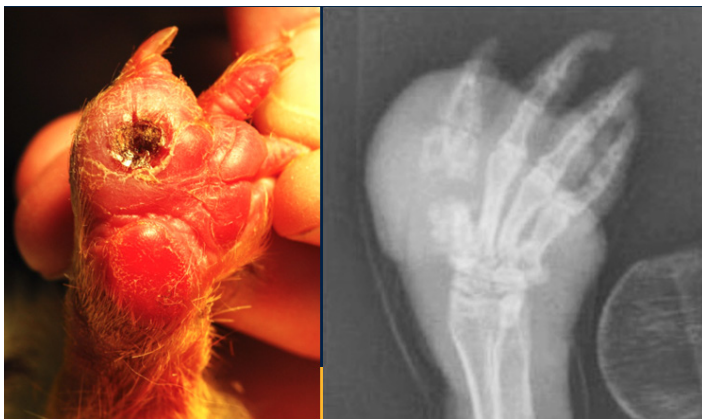
On imaging, articular changes may be seen such as the enlarged costochondral junctions of ribs and epiphysis, cortical thinning and demineralization of bones, and periosteal and epiphyseal separation of long bones.

Treatment mainly consists of supplementing with vitamin C at 50-100 mg/kg SC or IM q24h and then oral. Prevention should focus on dietary supplementation. Pellets specifically formulated for guinea pigs should be chosen that are fresh as vitamin C oxidizes quickly. **Vitamin C is unstable when exposed to light** and water supplementation is therefore not recommended as poorly effective in the long term. Vitamin C supplements such as **vitamin C tablets** can be given to guinea pigs or fruits and vegetables with a high content in vitamin C such as **oranges, red and yellow peppers, and strawberries.**

Pododermatitis

Pododermatitis is very common in pet guinea pigs and may affect any foot. **Predisposing factors include housing on wire cages, soiled bedding and poor hygiene, obesity, and concomitant musculoskeletal disorder.**

Lesions may range from mild erythema (grade I) to severe swelling and osteomyelitis (grade V). Grade II include erythema with gross swelling of the foot, grade III infection, ulceration and scabbing, and grade IV infection of deeper structure with abscesses and tendonitis. Radiographs may be taken to assess the presence of osteomyelitis and culture and sensitivity should be taken if ulcerated or during surgical debridement.



For grade I-II, treatment mainly consists of **correcting underlying issues and antibiotics/pain medications.** For higher grades, a combination of wound care,

surgical debridement, and amputation of osteomyelitic toes should be implemented. Wound management and bandages may have to be done for a long time. Prognosis is poor in severe cases.

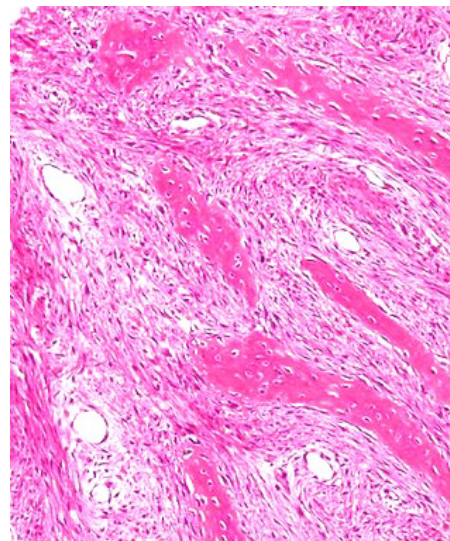
Below is a picture of a guinea pig foot after surgical debridement:



Fibrous Osteodystrophy

Fibrous osteodystrophy is seen in pet guinea pigs from times to times. It is mainly caused by **primary or secondary hyperparathyroidism (metabolic bone disease)**, likely caused by dietary deficiencies in calcium, improper dietary calcium-to-phosphorus ratio, vitamin D deficiency, calcium malabsorption or other calcium metabolic disturbances. **“Satin” guinea pigs are also predisposed** as up to a third of these animals can be affected, suggesting a genetic etiology.

There is increased osteoclastic activity in the bones and replacement by fibrous tissue causing swollen bones.



Clinical signs are related to the bone issues and include lameness, enlargement of the bones including the mandibles, lethargy, pain, and unwillingness to move. Pathologic fractures may also occur. The clinical signs may appear similar to scurvy.

The diagnosis is straightforward on radiographs with extensive changes observed in the bones. The calcium metabolism can be assessed by measuring plasma ionized calcium, phosphorus, Ca:P ratio, PTH, and vitamin D. ALP may also be elevated due to the increased osteoclastic activity

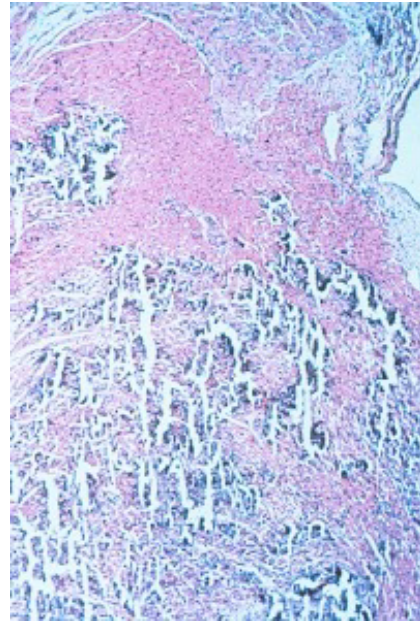
Treatment include **oral calcium supplementation, vitamin D supplementation if deficient and pain management for skeletal pain.**

The prognosis is guarded to poor and most guinea pigs don't survive more than 6 months after diagnosis.

Metastatic Mineralization

Metastatic mineralizations are seen on occasion in pet guinea pigs. They are caused by **hypervitaminosis D or excessive dietary phosphorus, magnesium, or calcium**. The excessive vitamin D comes from oversupplementation or by manufacturing issues of the pellets.

Lesions are **most often subclinical** but undetected muscle stiffness or renal dysfunction are possible. The mineralization are detected on radiographs and blood work may show an increase in calcium, vitamin D, and ALP. On pathology, extensive tissue mineralization is seen in multiple organs such as the kidney, stomach, heart, liver, uterus, colon and so on.



CHINCHILLA

Fractures

We will cover other cases of fractures in small mammals in the discussion session, but chinchillas are worth mentioning here as **fractures of the tibia** are exceedingly common.

Chinchillas have long straight tibia with little soft tissue coverage. They break easily resulting in transverse or short spiral fractures.

The initial treatment mainly consist of providing fluid therapy, nutritional support, and analgesia. External coaptation with a modified Robert-Jones bandage with or without a small splint is recommended. Surgical fixation is tricky as the tibial cortices are thin and fragile. Prognosis is guarded and post-operative complications are common. **Conservative management with bandage alone or a single IM pin in combination with a bandage** is an easy option. An external skeletal fixator or IM-ESF tie-in can also be attempted. Limb amputation is always a possibility in case of treatment failure.



The Ten Most Important Facts

1. Osteoarthritis and spondylosis are common in geriatric rabbits and manifest by stiffness of gait and lameness. Radiographic signs are similar to other species.
2. Splay leg in rabbits occur in young (hind limbs) and senior (front limbs) rabbits. Splinting or hobbling can be performed in young rabbits if caught early, but there is no easy fix for older bunnies.
3. Osteosarcoma is uncommon in rabbits and most reports are on the axial skeleton or mandible.
4. Ferret myofasciitis occurs in younger ferrets and manifests by acute lethargy, dysphagia, muscle pain and pyrexia.
5. In ferrets suffering from idiopathic myofasciitis, neutrophilic leukocytosis is seen on blood work and suppurative/pyogranulomatous inflammation is seen in atrophied muscles on biopsy or on post-mortem. Esophageal muscles are often affected as well.
6. Guinea pigs cannot make their own vitamin C and dietary deficiency will lead to scurvy, characterized by impaired collagen metabolism leading to joint issues (especially costochondral junctions), loose teeth, bleeding and a range of other issues. Oral supplementation should be made through vitamin C containing fruits and vegetables or in pellets and vitamin C tablets.
7. Pododermatitis is common in guinea pigs and may range from simple erythema to gross swelling, infection of deeper tissue and ultimately osteomyelitis. Treatment consists of husbandry modification, wound care, antibiotics, and surgical debridement.
8. Fibrous osteodystrophy is caused by metabolic bone disease and replacement of bony tissue by fibrous tissue. Satin guinea pigs are predisposed. Prognosis is guarded.
9. Metastatic mineralization in guinea pigs is caused by hypervitaminosis D or dietary imbalances in calcium, phosphorus, or magnesium. It is often subclinical.
10. Chinchillas are prone to fractures, especially on their tibia. Treatment is challenging and may include splinting, IM pin, or ESF fixation. Amputation is also an option when treatment has failed.